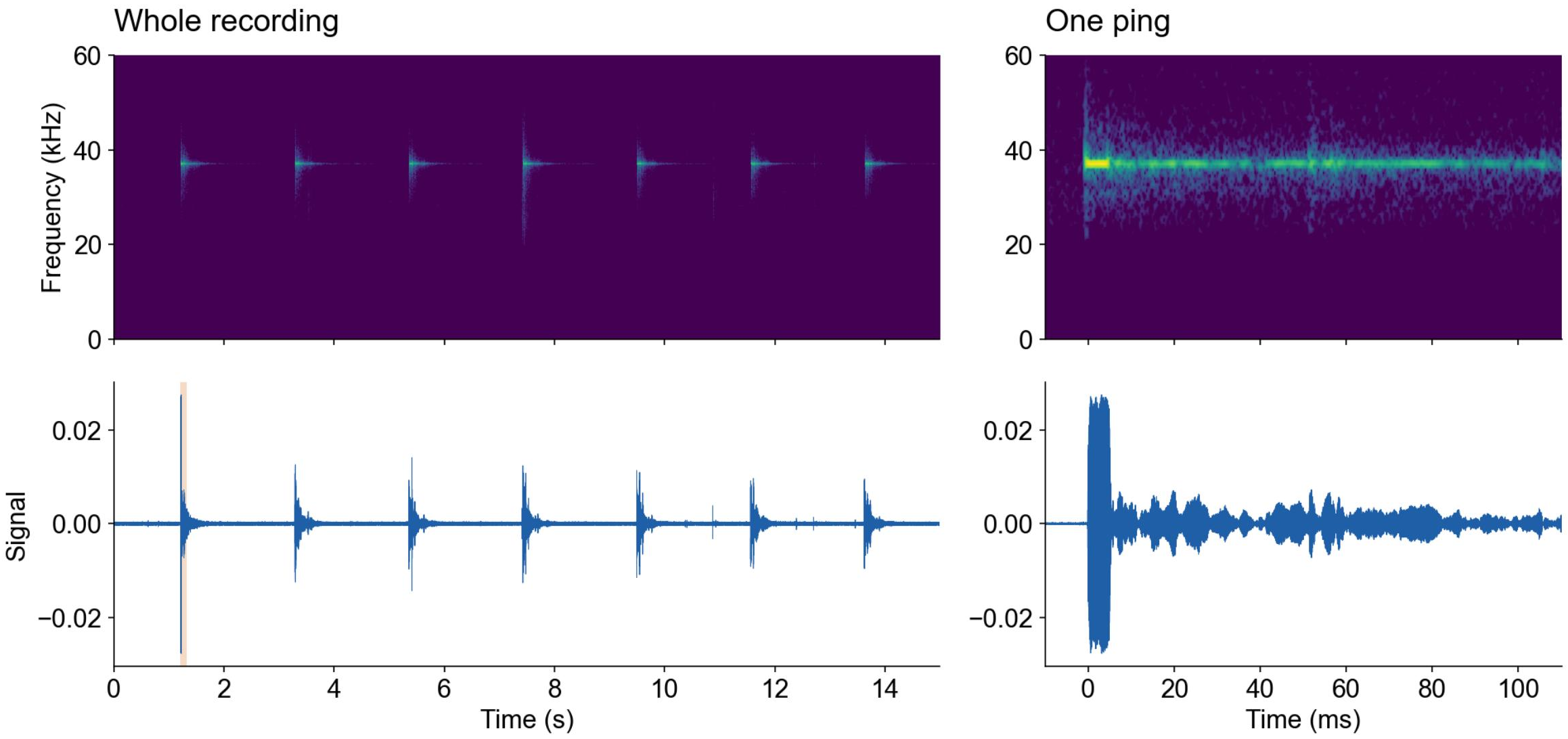
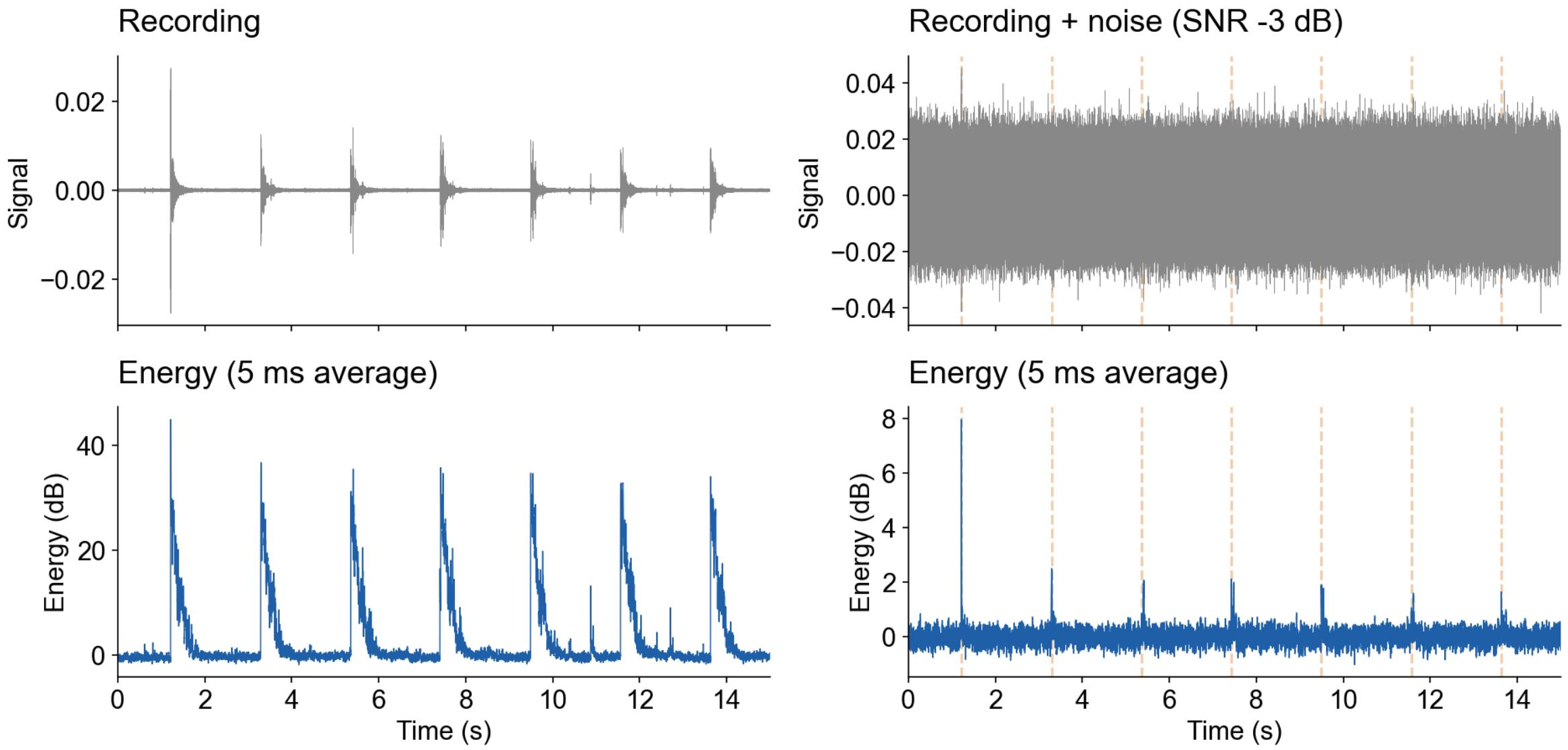
BTS challenge tutorial:
Signal design and DOA estimation

Recording


Whole recording
One ping
Frequency (kHz)
60
40
20
0
Signal
0.02
0.00
−0.02
0
2
4
6
8
10
12
14
Time (s)
0
20
40
60
80
100
Time (ms)

Energy detection

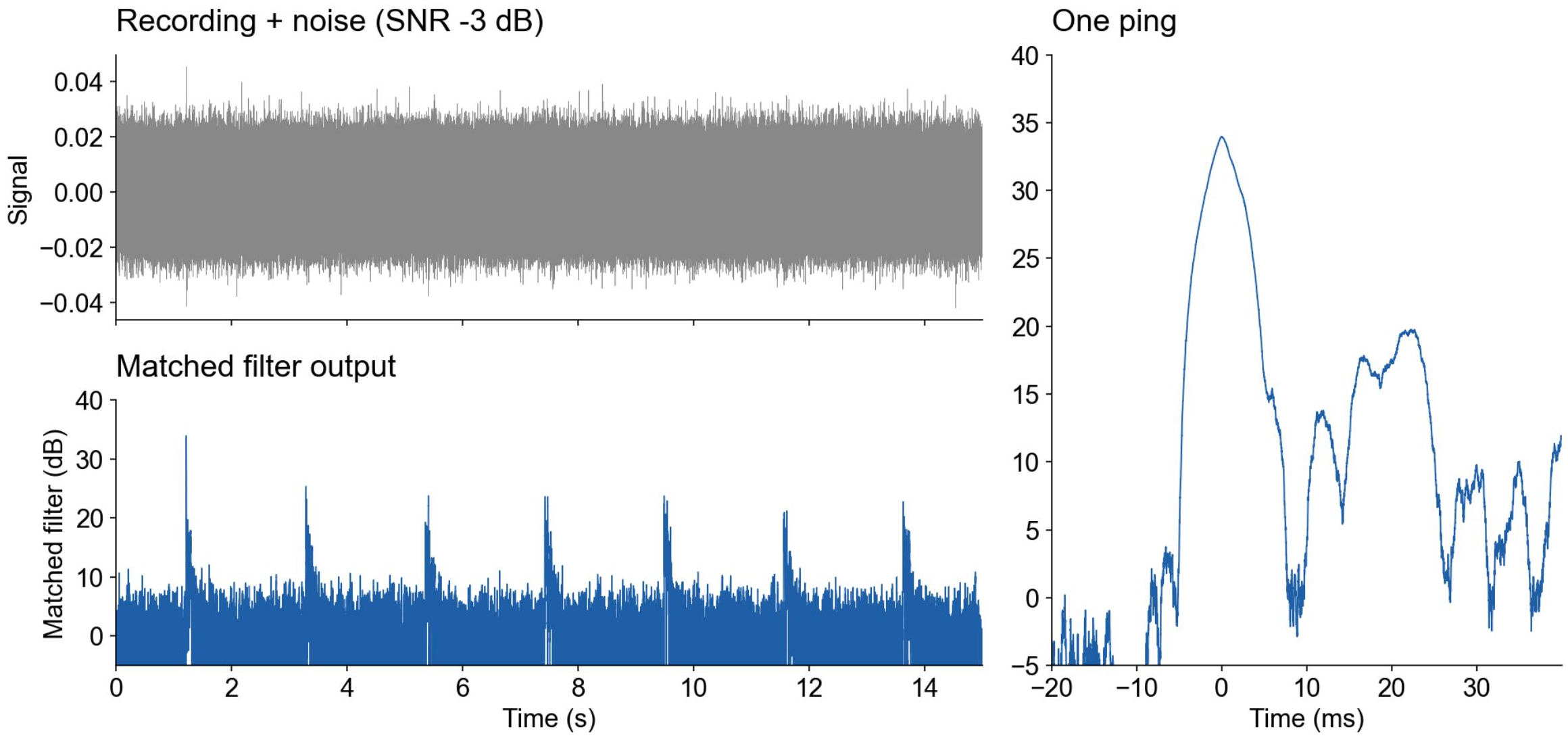


Matched filter

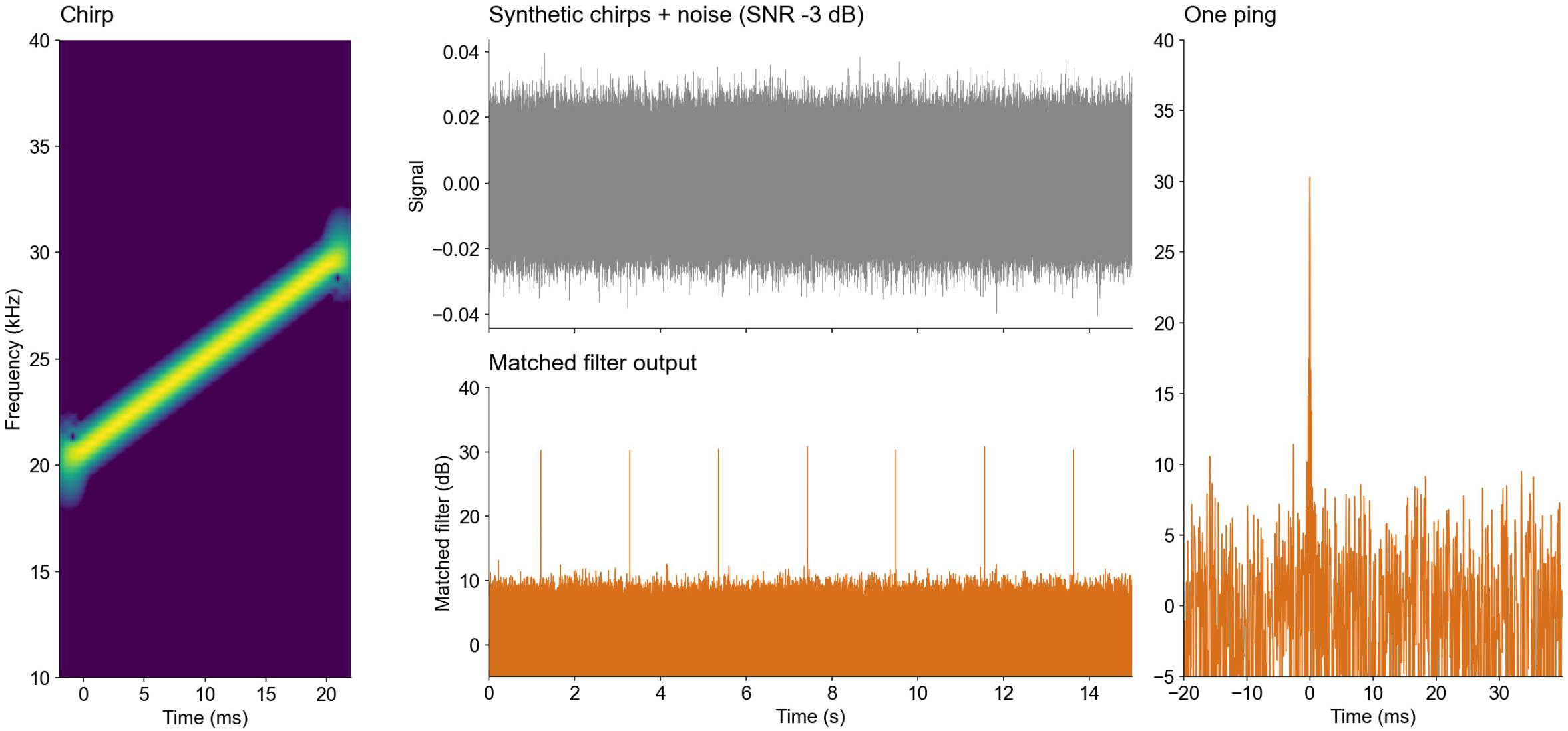
$$y(t) = \int x(t+u)\, s^*(u)\, du$$

x: received signal s: transmitted ping

Matched filter: CW ping



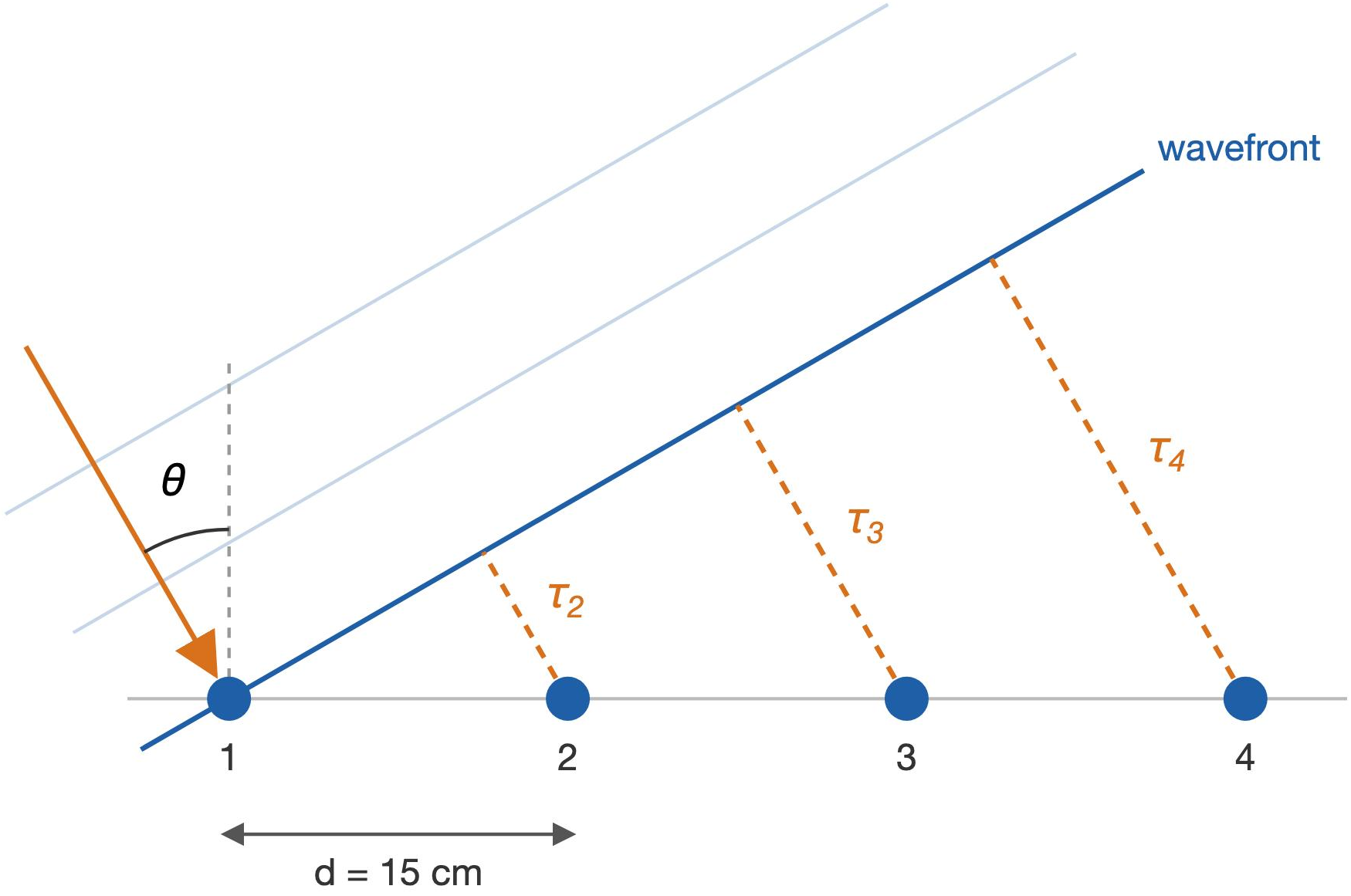
Matched filter: LFM chirp



Choice of signal

	CW ping	**LFM chirp**	**PN sequence**
Simplicity	Simple	Moderate	Moderate
Detection	Energy or matched filter	Matched filter	Matched filter
Timing accuracy	Poor (≈ ping length)	Good (≈ 1/B)	Good (≈ 1/B)
Multipath	Smeared	Resolved	Resolved
Doppler	Tolerant	Tolerant (small timing bias)	Sensitive

Time difference of arrival

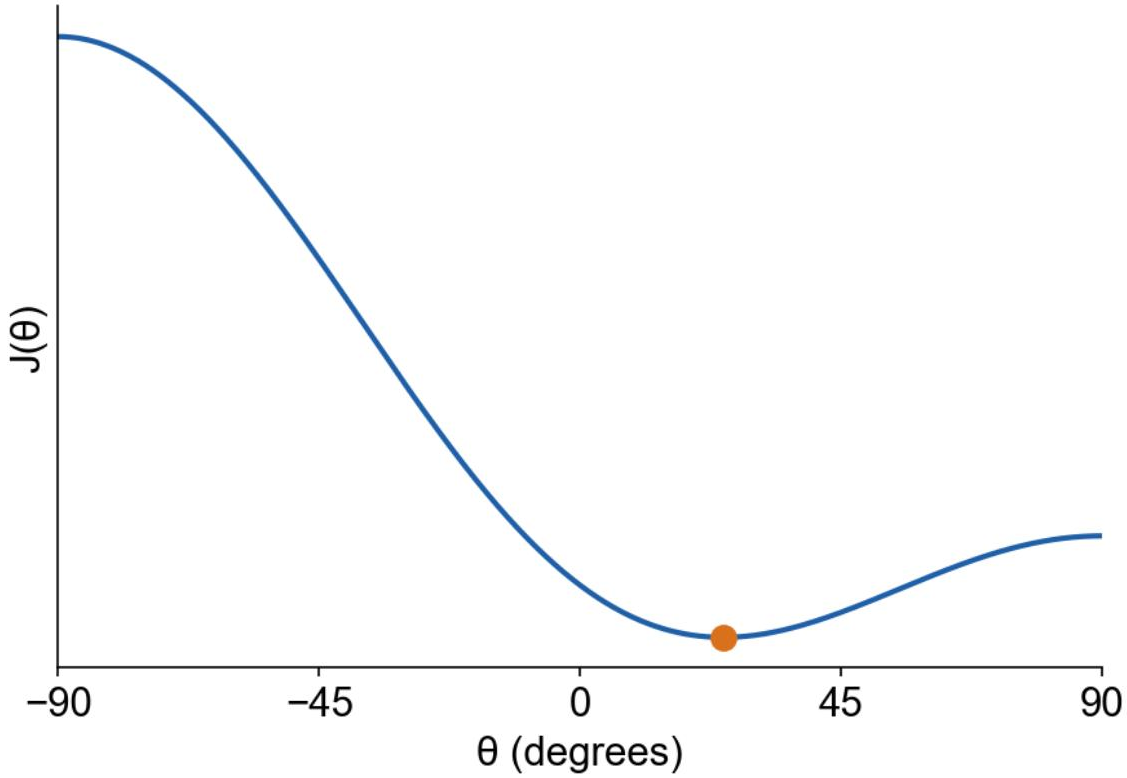


Estimating θ

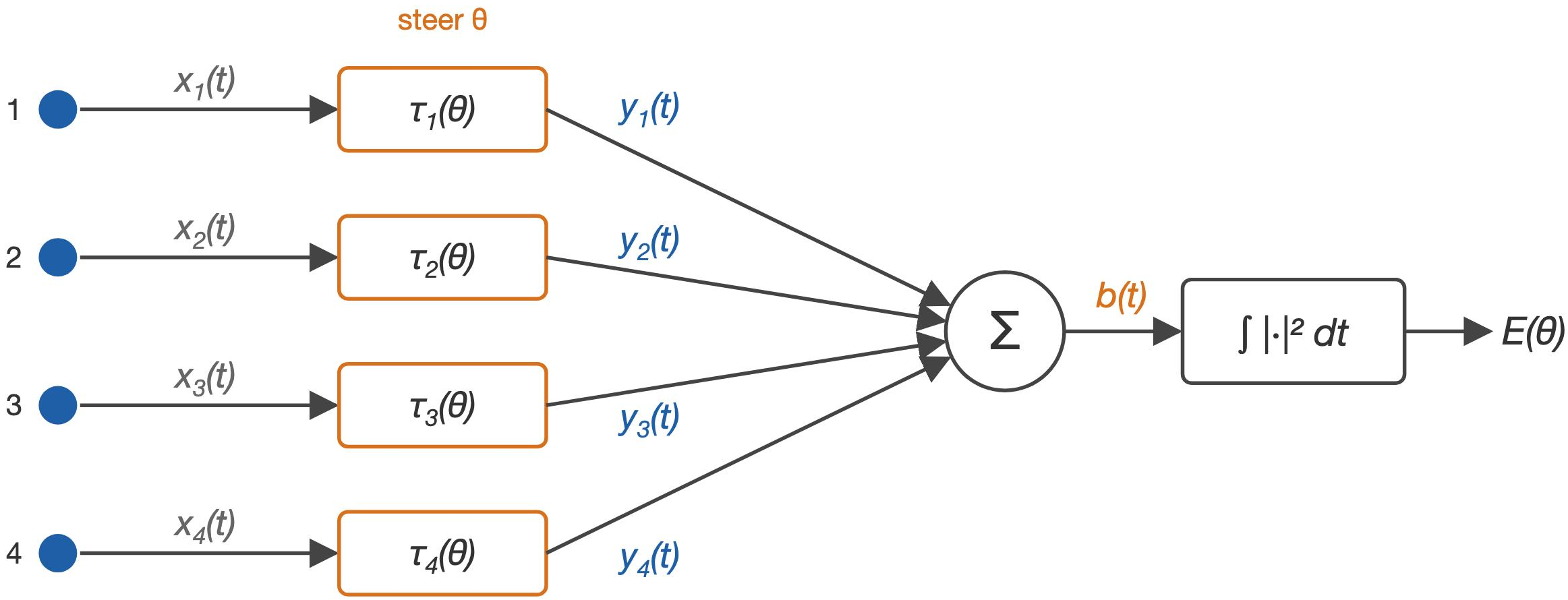
$$\tau_i(\theta) = \frac{(i-1)\, d \sin\theta}{c}$$

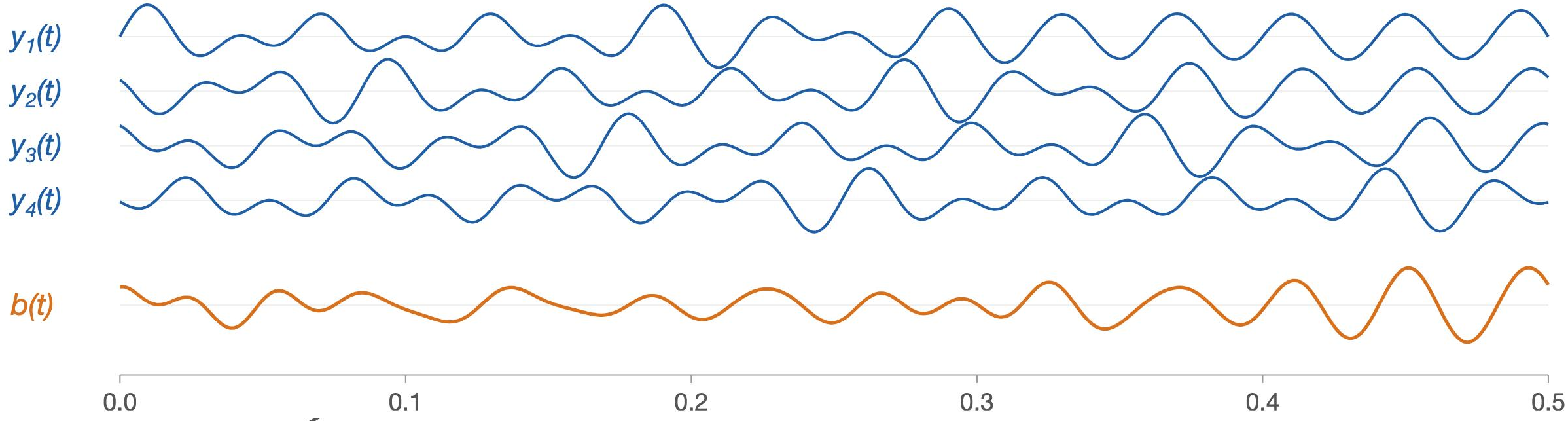
$$\hat{\theta} = \arg\min_{\theta} \sum_{i=2}^{4} \left(\hat{\tau}_i - \tau_i(\theta)\right)^2$$

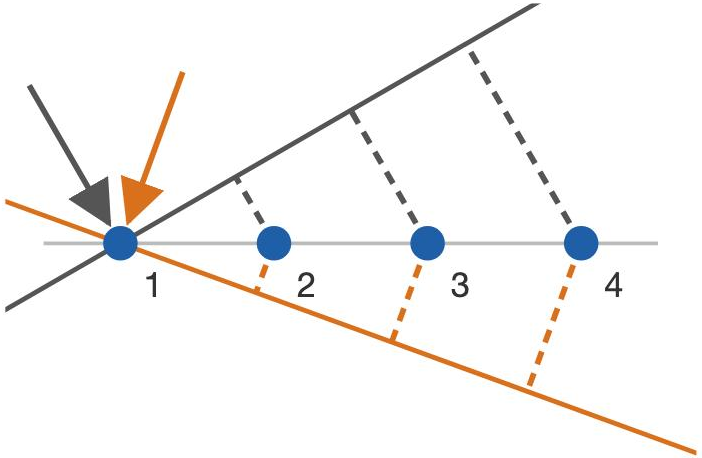
$\hat{\tau}_i$: measured delays



Delay-and-sum beamformer







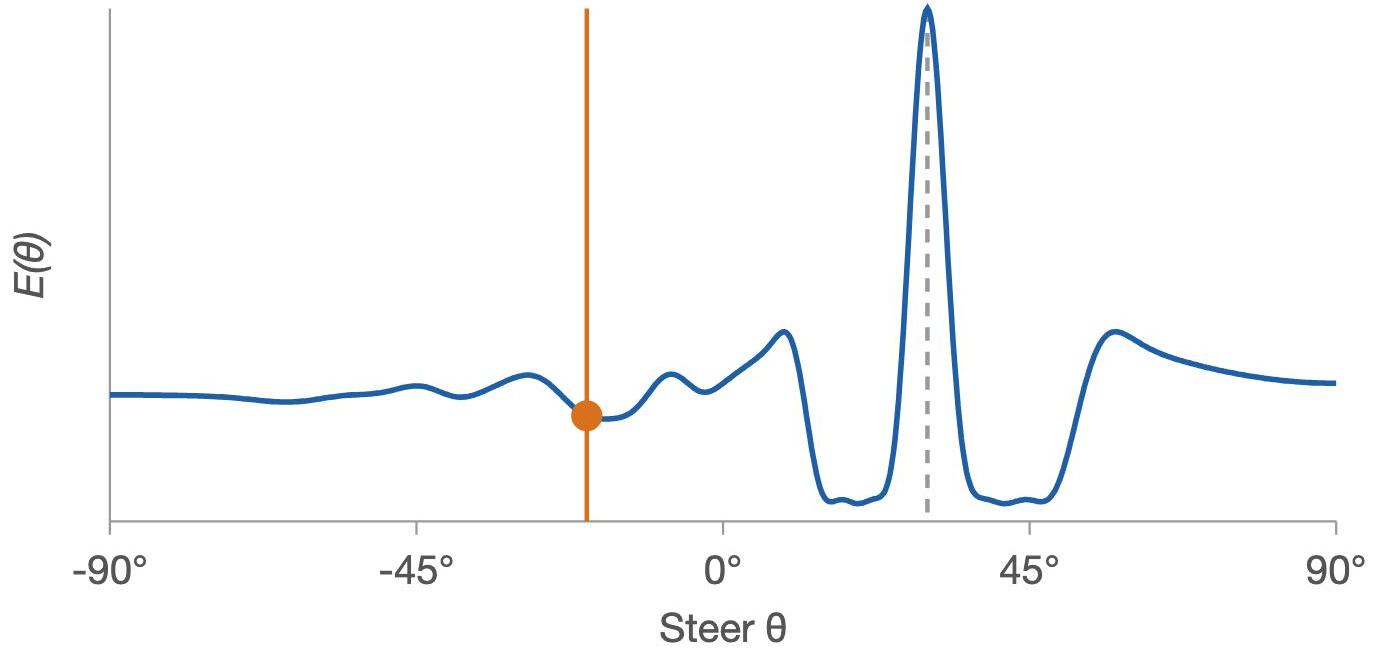
μs	actual	steer
τ_2	50	-34
τ_3	100	-68
τ_4	150	-103

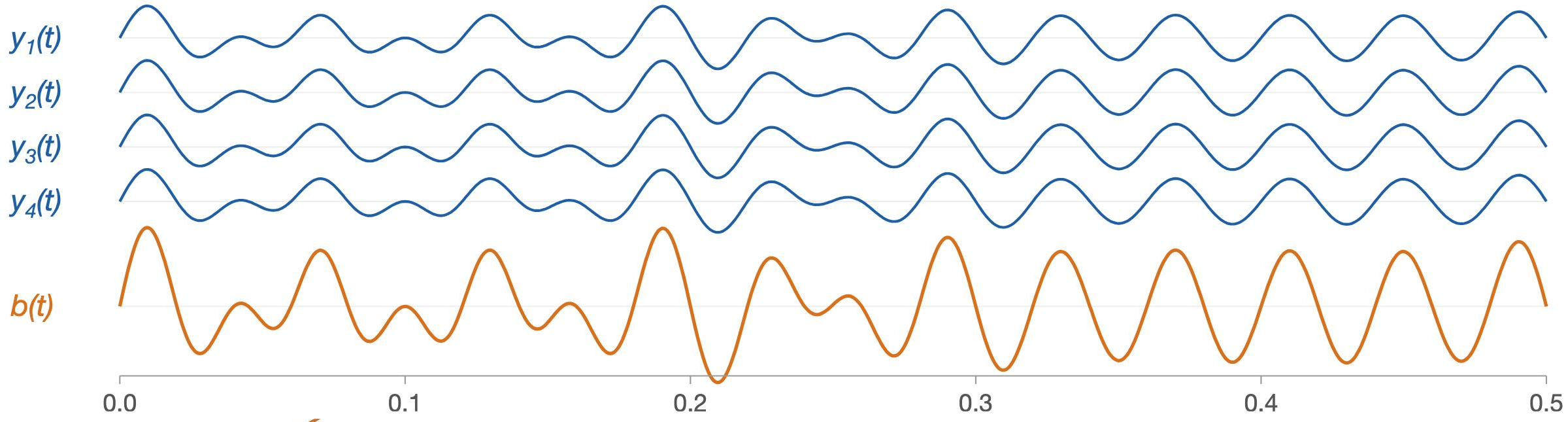
Arrival θ 30°

Steer θ -20°

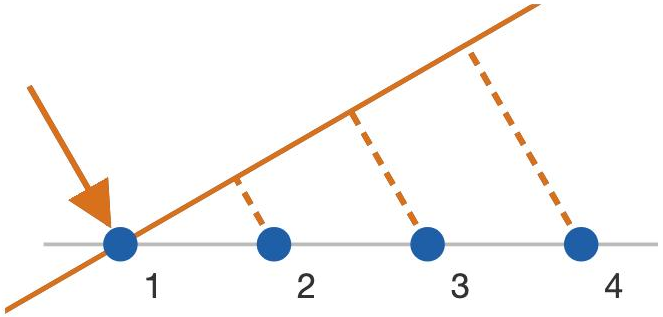
☑ Show E(θ) curve

E = 0.21

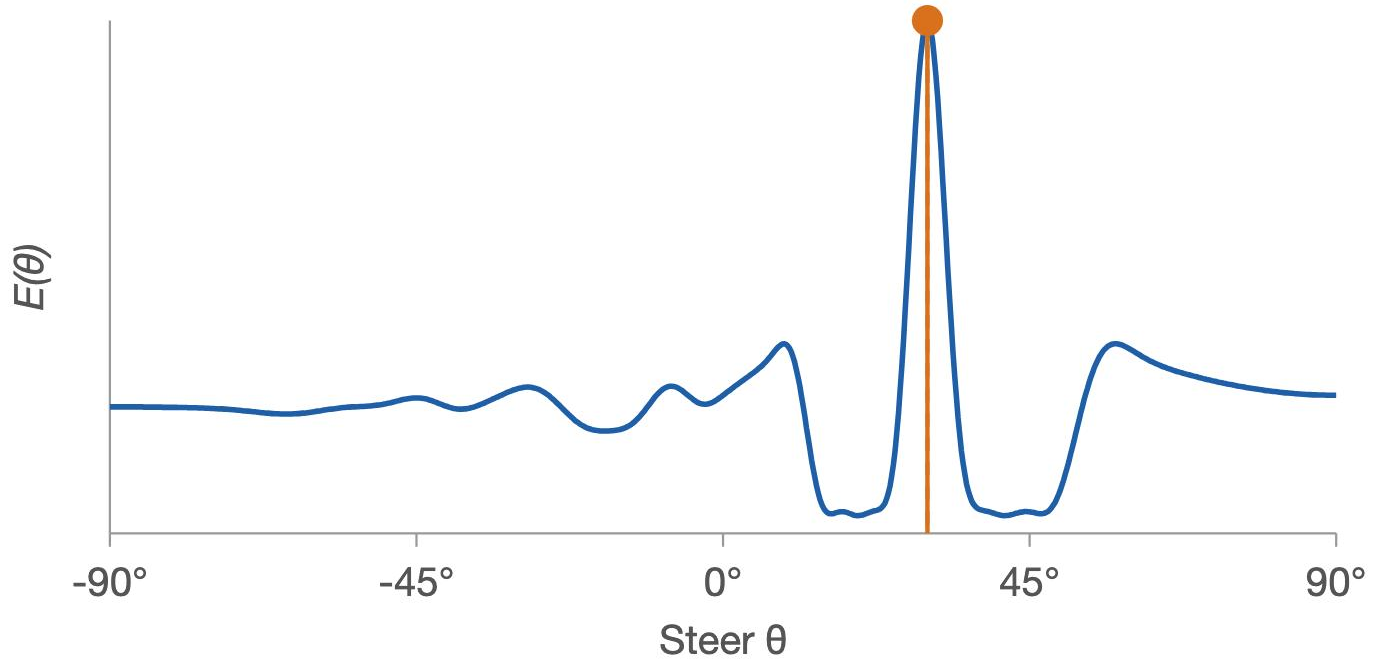




E = 1.00



μs	actual	steer
τ_2	50	50
τ_3	100	100
τ_4	150	150

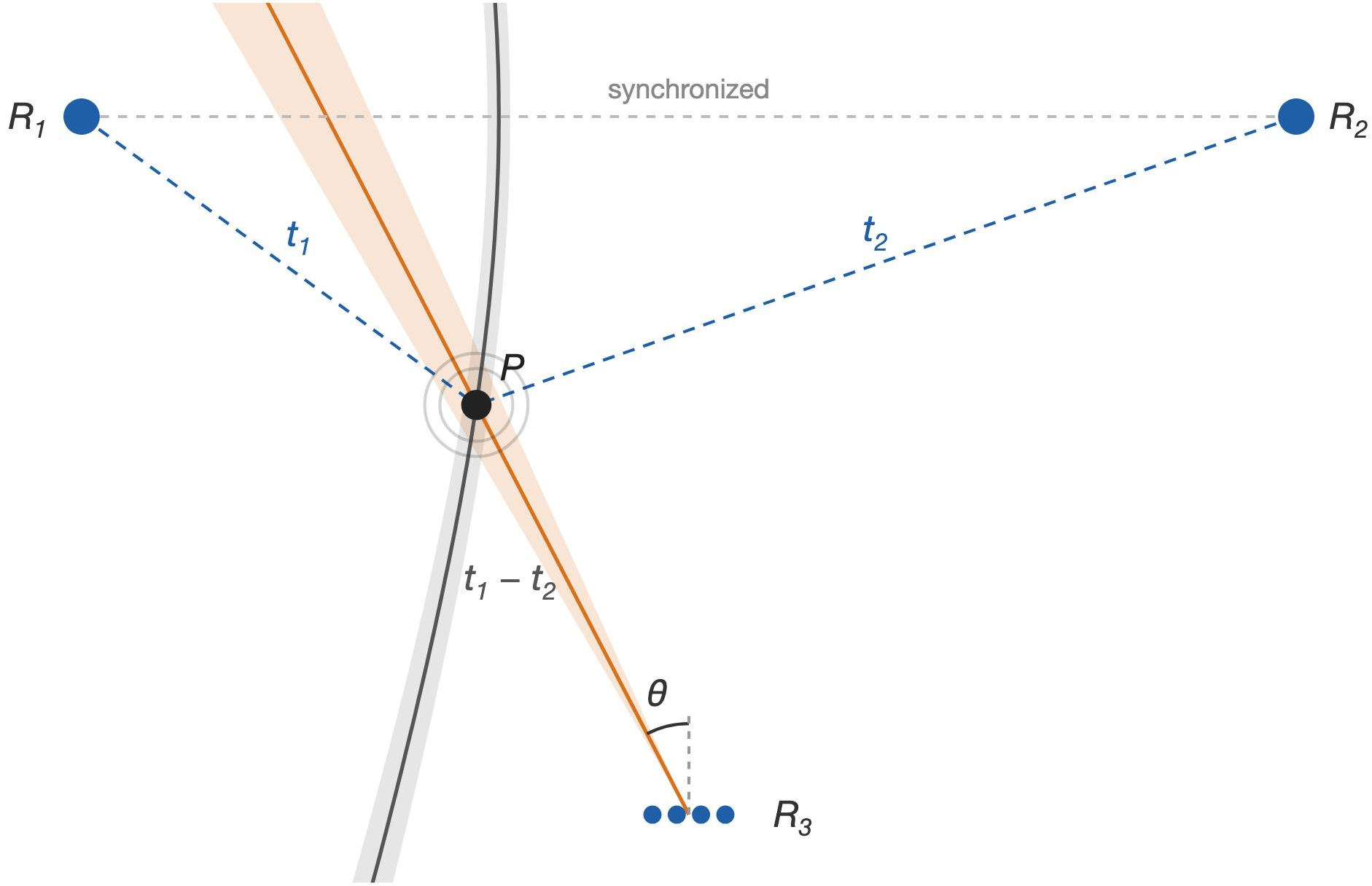


Arrival θ 30°

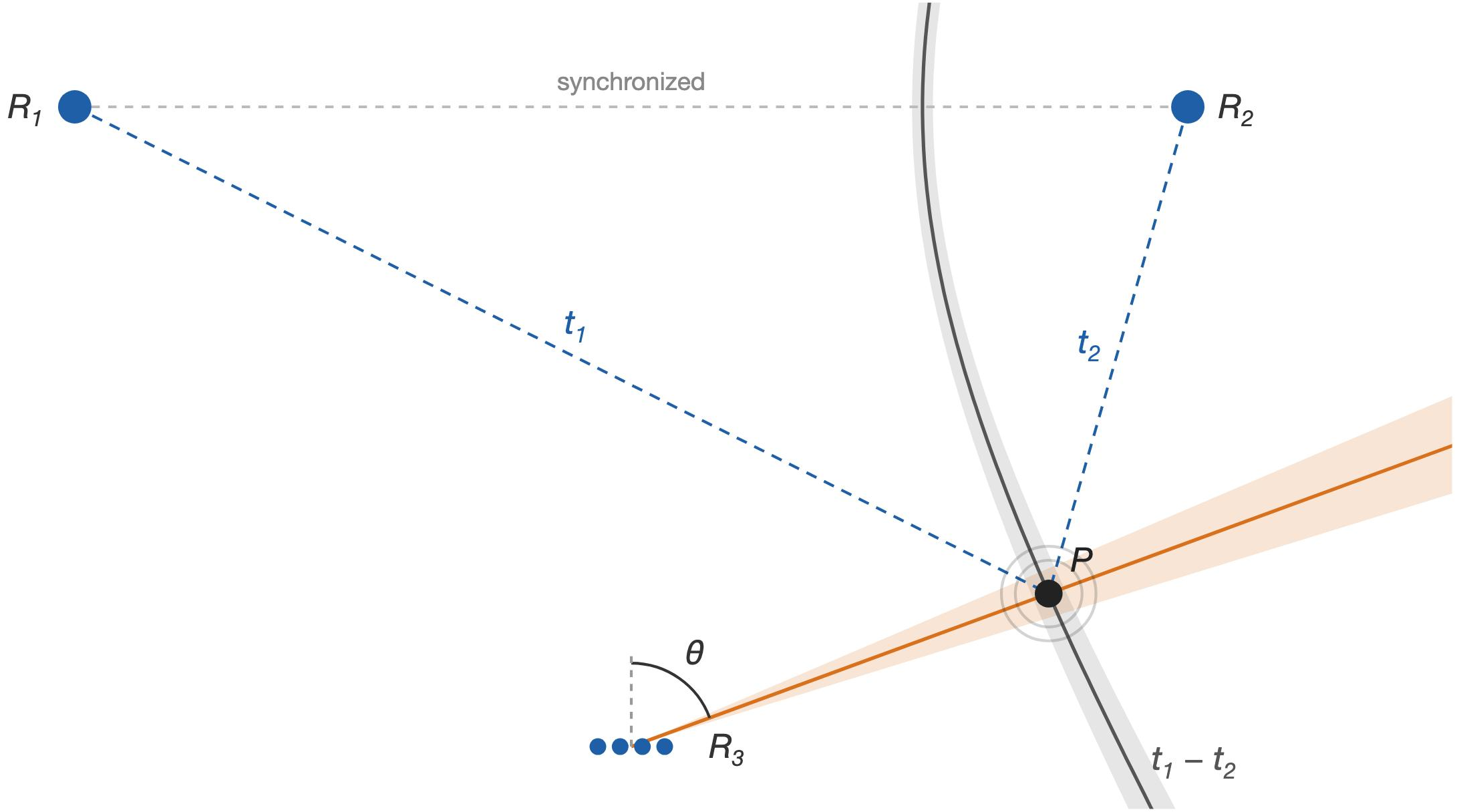
Steer θ 30°

☑ Show E(θ) curve

Positioning



Positioning



Tracking

- Motion model
- Kalman filter
- Particle filter
- Outlier rejection

